Lab: Full EDA on Netflix

Workshop 4 — Module 10

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Data Visualization for Data Scientists
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Module Roadmap

1. Apply the complete EDA pipeline to the Netflix dataset
2. Handle multi-valued columns (genres, countries) via explode
3. Separate Movie vs TV Show analysis (different target variables)
4. Build a nine-panel EDA dashboard
5. Workshop 4 review: three datasets, one framework

Learning Outcome

You will execute a complete, independent EDA on the Netflix dataset, demonstrating mastery of cleaning, missingness, outlier detection, correlation, and dashboard composition from Workshops 3–4.

The Netflix Dataset

Dataset Profile

Property	Value
Rows	8,807 titles
Movies / TV Shows	∼6,131 (70%) / ∼2,676 (30%)
Variables	12 (type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description)
Multi-valued	listed_in (genres), country, cast
Duration	Movies: minutes; TV Shows: seasons (different units!)
Missing	director (30%), country (10%), date_added (1%)

Pro-Tip

Netflix has two unique EDA challenges: (1) multi-valued columns require `separate_rows()` / `.explode()` before counting; (2) "duration" means minutes for movies but seasons for TV shows — never analyse them together without splitting.

Wrangling: The Netflix Cleaning Pipeline

R

```r
nf <- read_csv("netflix.csv") |>
  mutate(
    date_added = mdy(str_trim(
      date_added)),
    year_added = year(date_added)
        ,
    # Split duration by type
    dur_min = ifelse(
      type == "Movie",
      parse_number(duration), NA)
          ,
    seasons = ifelse(
      type == "TV Show",
      parse_number(duration), NA)
          ,
    # First genre / country
    primary_genre = str_trim(
      str_extract(listed_in,
        "^[^,]+")),
    primary_country = str_trim(
      str_extract(country,
        "^[^,]+")))

# Genre explosion (one row per
```

Python

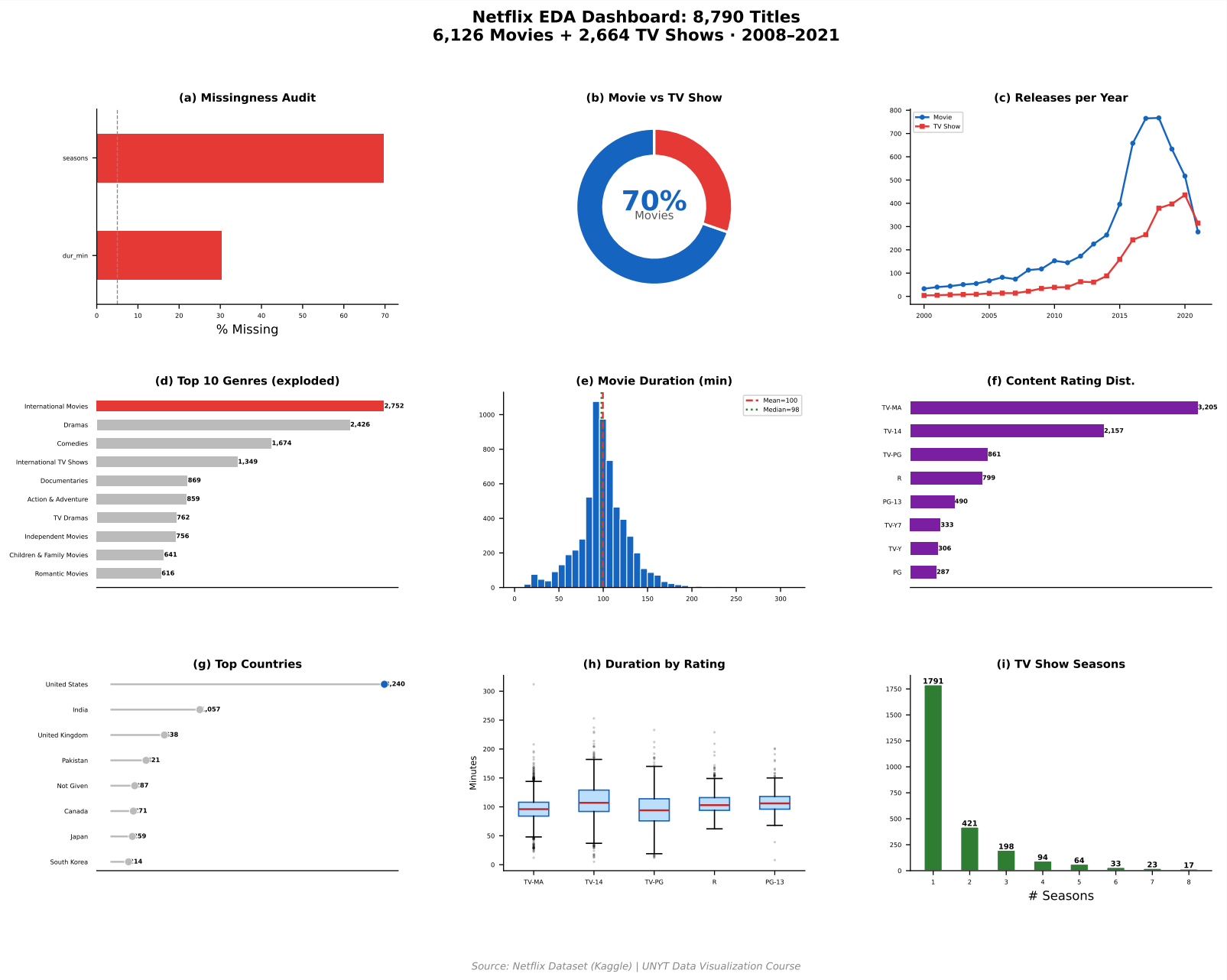
```python
nf = pd.read_csv("netflix.csv")
nf["date_added"] = pd.to_datetime
     (
    nf["date_added"].str.strip(),
    errors="coerce")
nf["year_added"] = (
    nf["date_added"].dt.year)

# Split duration by type
nf["dur_min"] = (nf
    .loc[nf["type"]=="Movie",
         "duration"]
    .str.extract(r"(\d+)")
    .astype(float))
nf["seasons"] = (nf
    .loc[nf["type"]=="TV Show",
         "duration"]
    .str.extract(r"(\d+)")
    .astype(float))

nf["primary_genre"] = (nf["
     listed_in"]
    .str.split(",").str[0].str.
         strip())
```

The Dashboard

Netflix: Nine-Panel EDA Dashboard



Reading the Dashboard

#	Panel	EDA Question	Key Insight
(a)	Missing % bar	Missingness audit	Director 30%, country 10%
(b)	Donut	Movie/TV split	70% Movie, 30% TV
(c)	Line by type	Content growth over time	TV surged post-2016
(d)	Genre bar	Which genres dominate?	Int'l Movies #1
(e)	Duration hist	Movie length shape	Near-normal, mean$\approx$100
(f)	Rating bar	Content rating dist.	TV-MA dominates
(g)	Country lollipop	Geographic origin	US 37%, India 14%
(h)	Duration by rating	Rating $\rightarrow$ length?	R-rated = longer
(i)	Seasons bar	TV season counts	72% are single-season

Key Findings

Four Key Findings

Netflix EDA: Four Key Findings

#	Finding	Evidence	Implication
1	**Netflix pivoted to TV Shows post-2016**	Movie additions peaked 2017 then declined; TV Shows grew steadily through 2021	**Content strategy shifted toward bingeable series**
2	**Director is 30% missing (MNAR for TV Shows)**	TV Shows often have no single director → structural missingness	**Analyse director only for Movies; encode as indicator**
3	**Movie duration is normally distributed (mean ≈ 100 min)**	Histogram + Q-Q near-normal; no heavy skew or multimodality	**Parametric methods OK for duration analysis**
4	**US dominates (37%), then India (14%)**	Lollipop shows long tail: top 2 countries = 51% of content	**Country is highly imbalanced; group rare countries**

Finding 1: The TV Pivot

- Movie additions peaked around 2017, then declined
- TV Show additions grew steadily through 2020–2021
- This mirrors Netflix's public strategy shift toward original series ("bingeable" content)
- **Implication**: any time-series analysis should account for this structural break. Pre-2016 and post-2016 are different populations.

Finding 2: Director Missingness is Structural (MAR)

- 30% of titles have no director listed
- TV Shows are far more likely to have missing directors than Movies (TV often has multiple directors per episode)
- This is **MAR**: missingness depends on the observed variable "type"
- **Implication**: analyse director only for Movies; for TV Shows, encode `has_director` as a binary feature or use the show's creator instead

Finding 3: Duration Is Near-Normal for Movies

- Mean $\approx$ 100 minutes, SD $\approx$ 28 minutes
- Histogram is approximately symmetric, slight right tail
- Q-Q plot follows the diagonal reasonably well
- **Implication**: parametric methods (t-tests, ANOVA) are appropriate for movie duration comparisons. No log transform needed (unlike Reviews in Google Play).

Finding 4: Extreme Geographic Concentration

- US produces 37% of Netflix content; India 14%
- Top 2 countries = 51% of all titles
- Long tail: 100+ countries each produce <1%
- **Implication**: group rare countries into "Other" for any categorical analysis. Use `fct_lump_n()` (R) or `.where(isin(top_n))` (Python).

Three Datasets Compared

One Framework, Three Domains

Three Datasets, One Framework: The EDA Pipeline is Domain-Agnostic

Dataset	n	Target	Missingness	Key Correlation	Domain
Google Play	10K apps	Rating (1–5)	MCAR (8%)	Reviews ↔ Rating (weak)	App marketplace
e-Car Loan	208K loans	Rate (%)	MNAR (53%)	FICO ↔ Rate (strong)	Consumer finance
Netflix	8.8K titles	Duration (min)	MAR (30% director)	Release year ↔ count	Streaming content

Workshop 4 Review

Workshop 4 Complete: Exploratory Data Analysis

Workshop 4 Complete: Exploratory Data Analysis

M01	**Tukey's EDA Philosophy** CDA vs EDA, detective, 4-plot
M02	**Wrangling for Viz (R)** dplyr verbs, pipe-to-plot
M03	**Wrangling for Viz (Python)** pandas chains, Rosetta stone
M04	**Missing Data Visualization** MCAR/MAR/MNAR, shadow, naniar
M05	**Outlier Detection** IQR, z-score, Mahalanobis, Cook's D
M06	**Correlation & Association** Anscombe, heatmap, pairs, ≠ causation
M07	**Case Study: Google Play** 9-panel dashboard, 4 findings
M08	**Case Study: e-Car Loan** Financial EDA, structural MNAR
M09	**Automation & Reproducibility** Auto-EDA, Quarto, parameterised
M10	**Lab: Full EDA on Netflix** Complete pipeline applied

The EDA Toolkit You Now Command

Capability	Tools	Module
Inspect & clean	dplyr, pandas, pipe chains	M02–M03
Diagnose missingness	naniar, missingno, shadow	M04
Detect outliers	IQR, z-score, Mahalanobis, Cook's D	M05
Measure association	Pearson, Spearman, Cramér's V	M06
Build dashboards	patchwork, GridSpec, 9-panel	M07–M08
Extract findings	Finding $\rightarrow$ Evidence $\rightarrow$ Implication	M07–M08
Automate & report	skimr, ydata-profiling, Quarto	M09

Workshop 4 Homework Summary

Due: Before Workshop 5
Perform a **complete EDA** on the Netflix dataset, following the seven-section report structure:

1. Data description & cleaning audit trail
2. Missingness audit (matrix + shadow analysis of director)
3. Univariate EDA (duration, rating, genre distributions)
4. Bivariate EDA (release year vs count, duration by rating)
5. Six-panel dashboard (patchwork + GridSpec)
6. Four numbered findings (Finding → Evidence → Implication)
7. Limitations and next steps

Total weight: 5% of course grade.

Next: Workshop 5 — Multivariate & Spatial Visualization